

Education			
Year	Degree/Certificate	Institute	CPI/%
Oct 2022 – Aug 2026	B.Tech in CS & IT	Ajay Kumar Garg Engineering College, Delhi NCR	7.25/10
2021 – 2022	Senior Secondary (XII)	New Era School, Delhi NCR	91.2%

Experience

Google Summer of Code 2025 Contributor | P4 Language Consortium, Stanford, CA (May 2025 – Sept 2025)

SplitDT Project: Scaling Stateful Decision Tree Algorithms in P4 P4, C++, Python, P4Runtime, Docker, Intel Tofino

- Architected a switch-native framework compiling ML decision trees directly into P4 data planes using Partitioned Decision Trees (PDTs) with TCAM-efficient Subtree IDs, working around hardware ASIC constraints on Intel Tofino silicon.
- Built a C++/P4Runtime data-path testing pipeline spanning Tofino hardware and Mininet software targets, improving deployment reliability by 30%.
- Optimized low-level SID recirculation logic to hit 10 Gbps line-rate throughput with up to 5x lower inference latency than control-plane inference.

Open Source Contributor - WasmEdge/WasmEdge (C/C++) (2024 – Present)

- Diagnosed and fixed a WASI clock bug in the C++ runtime causing 100% CPU usage on macOS using gdb and perf; merged to main runtime. (PR #4470)
- Built a standalone Alpine-base Docker image build pipeline for the runtime. (PR #4466)
- Upgraded the base Alpine image to 3.23 and restored static LLD linker support. (PR #4440)

- facebook/bpfilter)

- build: update clang-tidy brace config, fix opts.c/set.c, and NOLINT dump.c (PR #507)

Open Source Contributor - Project-HAMi/HAMi (CNCF Kubernetes GPU Device-Sharing) (2024 – Present)

- Fixed scheduler and device-plugin bugs in HAMi, a CNCF project for GPU/device resource sharing and isolation across Kubernetes clusters - 5 PRs merged.

Projects

KNIC v1 | C, Linux Kernel, PCI, DMA, NAPI, QEMU, DPDK

- Built a clean-room Intel 82540EM/e1000-compatible PCI Ethernet driver in C: PCI probe/remove, BAR0 MMIO, coherent and streaming DMA rings, interrupt-driven NAPI, TX backpressure, ethtool telemetry, and watchdog/reset recovery.
- Qualified it under KFENCE and sparse with staged fault injection: 646,400 packets at zero loss, 1,000 interface cycles, 100 rebinds, and 100 reset-under-traffic cycles on reproducible QEMU/HVF automation.
- Added a W5500 SPI driver path and a gated DPDK L2 forwarder; validated partial-TX cleanup and signal-safe shutdown against the net_pcap virtual PMD.

ebpf-mcp-tracer | Python, bpftool, eBPF, Linux

- Secure server exposing controlled Linux kernel tracing to LLM agents: strict probe allowlist, dry-run validation, and hard timeouts so agents can debug system bottlenecks without raw unsafe eBPF access.

Los-Tecnicos | Go, TypeScript, Rust, Soroban, Raspberry Pi, ESP32

- Decentralized energy-grid system with a Raspberry Pi/ESP32 hardware mesh; built the Go backend and 4 Soroban smart contracts for SW/HW bring-up; won Asia-Pacific Web3 bounties, recognized by the Government of India.

Technical Skills

- Languages: C, C++, Python, Go, P4, JavaScript, TypeScript, Rust
- Systems / Networking: Linux (x86_64, ARM64, AMD64), eBPF, gRPC, Protobufs, Scapy, socket-level networking
- Debugging / Profiling: gdb, perf, pprof, htop
- Tools / Platforms: Docker, CI/CD, GitHub, GitLab, UTM
- Databases: SQL, PostgreSQL, MySQL, MongoDB, Redis, GraphQL

Publications

- R. Bharadwaj, S. Jha, V. Malyan, R. Gupta, "Trusture: A Blockchain-Based Decentralized Framework for Secure, Transparent, and Auditable NGO Transactions," 2026 ICCSC, Ghaziabad, India, Feb. 2026. DOI: 10.1109/ICCSC67078.2026.11468597

Leadership

- Technical Writer, Cloud Native Community Group, New Delhi (May 2024 – Dec 2024)
 - Led 6+ writers and 4 designers on documentation and event content; social media revamp lifted engagement 60%.
- Co-Organizer & Anchor, Google Developer Groups & GDSC WOW 2024, Delhi NCR (Nov 2023 – Nov 2025)
 - Co-led 10+ tech events; anchored GDSC WOW at IIIT Delhi (800+ live audience, 31+ crew).

Achievements & Certifications

- 2x AKTU State TT Gold (2023, 2024); Robotics Olympiad Bronze, Thailand (2019); NASA, NPTEL & ISRO certifications